

AI-powered Last-Mile Delivery Exception Handling Automation

Advanced Agentic AI Solutions



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- Executive Summary
- Business Problem Overview and Solution Approach
- Dataset Overview and Model Initialization
- Defining Tools
- Defining the Multi-Agent System Architecture
- Test Cases Execution, Evaluation, and Observability

Executive Summary - Conclusions

1 Deterministic guardrails are more reliable for rule-based decisions

- ❖ Deterministic noise filtering and escalation enforcement achieved 100% accuracy.
- ❖ Embedding unambiguous business rules in code rather than relying on LLM prompt compliance proved to be the highest-confidence design pattern.

2 The system handles clear-cut exception types with high accuracy and consistency

- ❖ Damaged packages, refused deliveries, perishable weather delays, and standard failed attempts all resolved correctly on the first pass with zero revisions and coherence scores of 4–5 out of 5.

3 Resolution accuracy degrades at playbook ambiguity boundaries

- ❖ The two failures involve scenarios where the playbook supports multiple valid interpretations.
- ❖ The system's eventual escalation on these cases is the operationally safe outcome; these are judgment calls that would split human operators as well.

Executive Summary - Conclusions

4

The Critic Agent's revision loop is a double-edged sword

- ❖ The Critic catches genuine errors, but on ambiguous inputs it can push the Resolution Agent away from a correct initial answer, increasing latency and producing a worse decision

5

Communication quality was consistently high across all exception cases

- ❖ Tone, channel format, message content, and service credit acknowledgment matched expectations across all exception cases with correct PII handling throughout.

6

The security architecture operated transparently without impacting results

- ❖ Agent state isolation, PII redaction, and prompt injection scanning were active across all test cases with no false positives and no measurable latency impact.

Executive Summary - Business Recommendations

Route to a human supervisor when the Critic and Resolution Agent oscillate, rather than exhausting retries and risking overriding a correct initial decision.

Auto-escalate ambiguous cases instead of looping

Measure cost-per-exception reduction for business case validation

With last-mile expenses at up to 53% of total shipping costs, measure cost-per-exception and time-to-resolution before and after deployment.

1

2

3

4

Expand to larger labeled shipments covering seasonal variation and diverse driver note phrasing for statistically significant measurement.

Expand the test case dataset and integrate real-world delivery feeds

Add a feedback loop from supervisor resolutions to improve the playbook and prompts

Feed supervisor resolutions back into ground truth and agent prompts to reduce future escalation volume on recurring patterns.

Business Problem Overview

Business Context

In last-mile logistics, roughly 10% of all shipments encounter delivery exceptions - failed attempts, address mismatches, damaged packages, refused deliveries, and weather delays. Each failed delivery costs in direct reattempt expenses, while the downstream impact is far steeper: consumers stop shopping with a retailer after just 2-3 failed deliveries, and supply chain disruptions cost companies millions per year.

Despite these costs, most organizations still handle exception triage and resolution manually. Operations staff read through status logs and messy driver notes, cross-reference customer profiles and internal playbooks, decide on a resolution action, and draft customer notifications - all under time pressure. Teams rely on support tickets, tribal knowledge, and rigid rule engines that cannot handle the nuanced, multi-factor judgment calls that real exceptions demand. A VIP customer with a perishable package and a broken gate code on a second attempt requires a fundamentally different resolution than a standard customer's first failed attempt on a non-perishable item - but manual processes treat them with the same slow, inconsistent workflow.

Business Problem Overview

Business Context

Key KPIs affected include:

- Exception resolution time (constrained by manual triage throughput)
- Escalation accuracy (missed or unnecessary supervisor escalations)
- Customer communication quality and personalization
- Cost per exception (reattempt costs, spoilage write-offs, unnecessary truck rolls)
- Customer retention and lifetime value

If unaddressed, logistics providers face rising exception-handling costs, inconsistent service quality, preventable customer churn, and loss of competitive ground to players who are already investing in AI-powered exception detection and response automation.

By applying an AI-powered multi-agent system that ingests delivery logs, cross-references customer profiles and locker availability, retrieves resolution policy from an operational playbook, and generates validated decisions with personalized customer communications, logistics providers can automate the full exception-handling pipeline from detection through resolution, with built-in quality validation and supervisor escalation where policy requires it.

Business Problem Overview

Objective

The objective is to build a POC of an AI-powered multi-agent delivery exception handling system for the last-mile delivery operations of a mid-sized retailer that:

- uses the operator's available data to process exceptions end-to-end,
- ingests raw delivery status logs, including noisy, duplicated, and multi-row shipment events, and correctly identifies actionable exceptions versus routine operational noise,
- decides the appropriate resolution action (reschedule, reroute to locker, replace, or return to sender) by reasoning over playbook rules, customer context, package constraints, and locker eligibility, while respecting operational policies around perishable handling, fragile thresholds, and locker capacity.

Business Problem Overview

Objective

- escalates to a human supervisor when policy requires it and avoids unnecessary escalations that waste supervisor capacity,
- generates personalized customer notifications with tone, channel, and content calibrated to customer tier and exception severity, and
- produces auditable decisions with step-by-step rationale, critical validation traces, and system evaluation metrics across five dimensions: task completion rate, escalation accuracy, tool call accuracy, reasoning trajectory coherence, and end-to-end latency.

The end goal is to demonstrate measurable accuracy and consistency on curated exception scenarios sufficient to justify scaling the approach to real-time operations with live delivery feeds, broader exception taxonomies, and multi-region deployment.

Dataset Overview

- A structured backend database (`customers.db`) containing customer profiles and locker locations

Database Table Name	Database Table Content
customers	Contains PII, service tiers, communication preferences, recent service exceptions, and active credit balances
lockers	Contains physical storage locations, detailing their addresses, real-time capacity status, operating windows, and the maximum package dimensions they can accommodate

- A PDF file (`exception_resolution_playbook.pdf`) containing an internal operations manual for human dispatchers in natural language, containing exception handling procedures, escalation matrix, locker reroute eligibility rules, and customer communication guidelines.

Dataset Overview

- CSV files include the delivery status logs with free-text driver notes across 10 shipments, and the ground truth labels.

CSV File Name	CSV File Content
Delivery Logs	Contains 13 events across 10 unique shipments, documenting status transitions, package specifications, customer associations, and attempt counts to monitor the lifecycle of deliveries.
Ground Truth	Contains a 1:1 mapping to delivery events, defining exception classifications, prescribed resolutions, and communication protocols justified by systematic reasoning against business logic and service playbooks.

Model Initialization

- We set up separate LLM instances for output generation and output validation.

Generation Model

GPT 4o-mini

Temperature: 0

Validation Model

GPT 4o

Temperature: 0

Defining Tools

1

`read_delivery_logs`

Reads all delivery log rows from the CSV file into a list of dictionaries.

2

`lookup_customer_profile`

Retrieves customer data by ID, using a conditional `include_pii` flag to restrict PII access only to the Communication Agent.

3

`check_locker_availability`

Filters local lockers by zip code and evaluates them against constraints to return a list of valid locations with clear justification for each.

4

`search_playbook`

Identifies relevant knowledge chunks from the vector store and returns their text alongside metadata for citation and traceability.

5

`check_escalation_rules`

Evaluates hard-coded business rules and regex-parsed delay data to categorize shipments as automatic or discretionary triggers (no LLM intervention).

Multi-Agent System Architecture - Agent State

UnifiedAgentState: A single, global data structure that carries every field across the entire LangGraph pipeline, serving as the "source of truth" for the duration of the process.

Sub-Agent View: A security layer using typed dataclasses and utility functions (`project_into` and `merge_back`) to strictly limit each agent's read/write access to only their designated fields.

We define a multi-agent system with four sub-agents

1

Router Agent

2

Resolution Agent

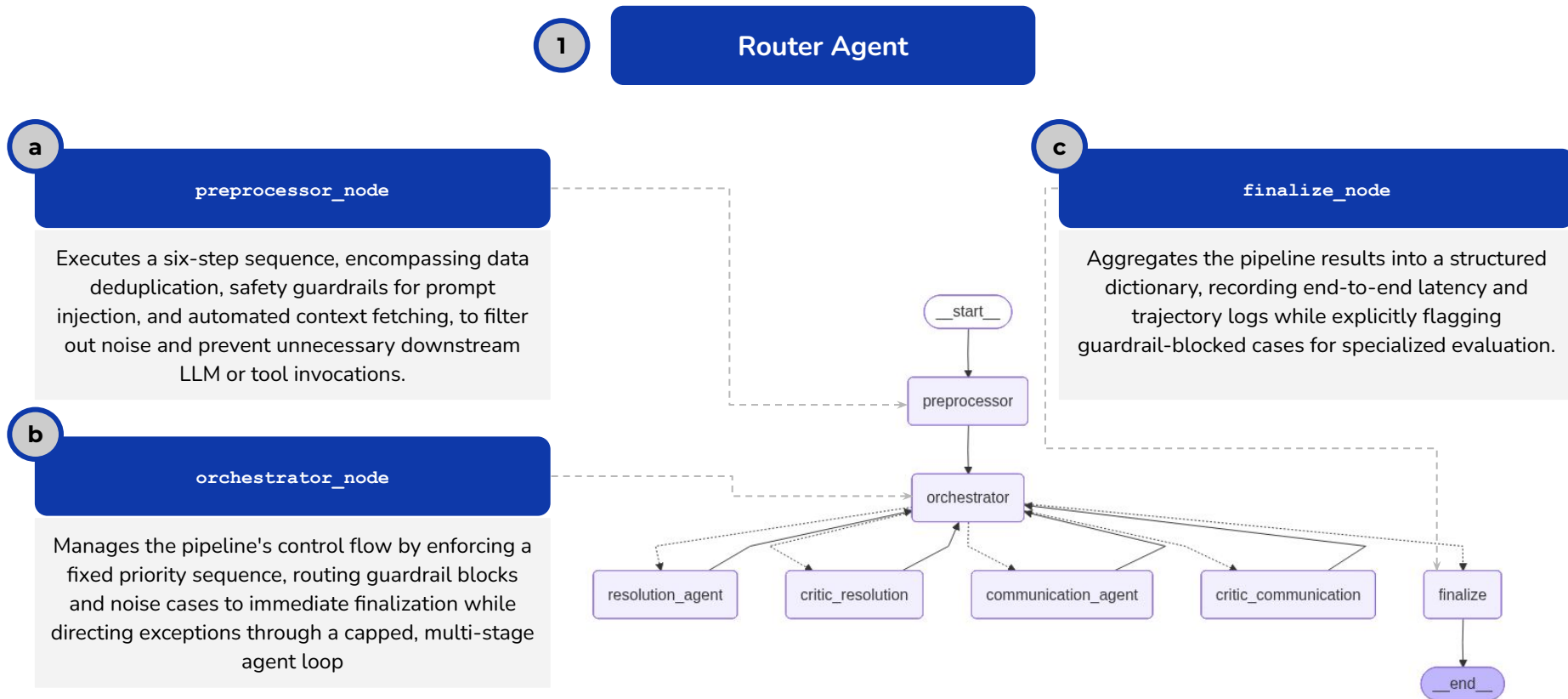
3

Communication Agent

4

Critic Agent

Multi-Agent System Architecture - Sub Agents



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Multi-Agent System Architecture - Sub Agents

2

Resolution Agent

a

resolution_agent_node

Manages the core decision-making logic by processing non-PII context and Critic feedback through a structured LLM prompt, utilizing a Pydantic-validated retry loop that defaults to human-escalated rescheduling if validation fails.

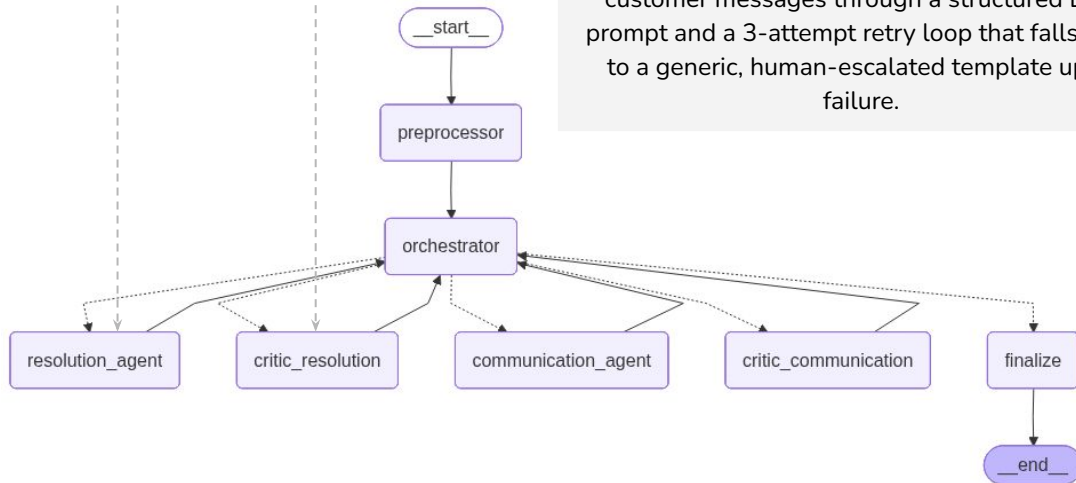
3

Communication Agent

a

communication_agent_node

Acts as the sole PII-privileged node, personalizing customer messages through a structured LLM prompt and a 3-attempt retry loop that falls back to a generic, human-escalated template upon failure.



Multi-Agent System Architecture - Sub Agents

4

Critic Agent

a

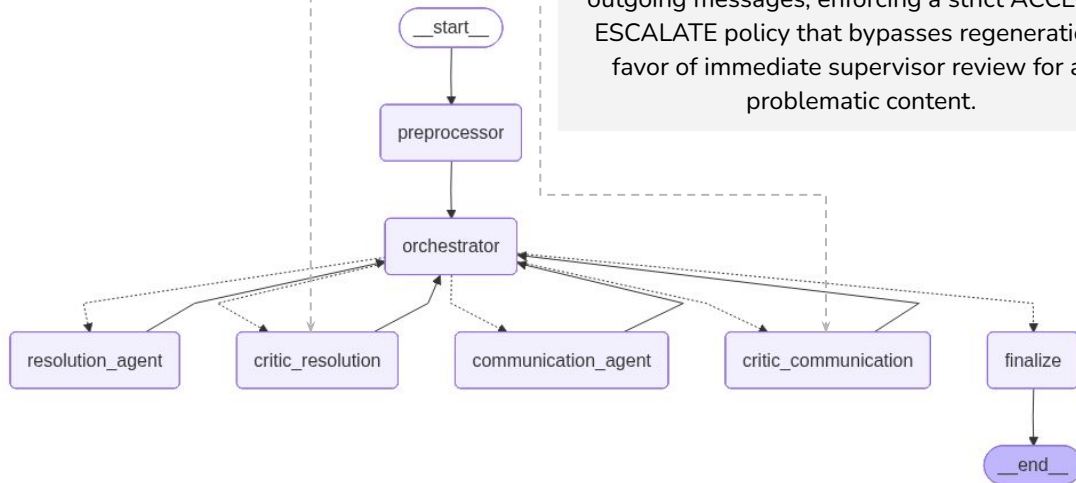
`critic_resolution_node`

Validates agent decisions without PII access, triggering either a global escalation flag or a revision loop that provides specific rationale feedback to the Resolution Agent until the retry limit is reached.

b

`critic_communication_node`

Performs a final, PII-blind quality check on outgoing messages, enforcing a strict ACCEPT or ESCALATE policy that bypasses regeneration in favor of immediate supervisor review for any problematic content.



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Test Cases Evaluation Metrics

1

Task Completion Rate

Calculates a binary success metric for each shipment, requiring a perfect match across exception classification, resolution action, and communication tone (with "N/A" exclusions for noise cases) to count as a completed task.

2

Escalation Accuracy

Measures the system's boolean escalation logic against ground truth labels, excluding "N/A" noise cases to focus specifically on the precision of exception handling.

3

Tool Call Accuracy

Validates system integrity by verifying that the correct execution path was taken - zero tool calls for short-circuited noise, a full suite of six tools for exceptions, and a five-tool subset (excl.Communication Agent) for non-exceptions.

4

Reasoning Trajectory Coherence

Evaluates the logical flow of a test case lifecycle by processing a PII-redacted trace of agent outputs, critic decisions, and revision counts through an LLM to extract a structured coherence score and justification.

5

Latency

Measures the total time elapsed from the moment the Preprocessor Node receives a shipment to the moment the Finalize Node emits the structured output dictionary.

Test Cases Evaluation Summary

- The system achieved 100% tool call accuracy and 87.5% escalation accuracy.
- The two task completion failures ("SHP-003" and "SHP-005") stem from ambiguous playbook interpretation and infeasible locker constraints, respectively, not from tool or routing errors.
- The average coherence score of 4.30/5.0 indicates that agent reasoning is consistently sound.

Evaluation Metric	Value
Task Completion Rate	80.00%
Escalation Accuracy	87.50%
Tool Call Accuracy	100.00%
Reasoning Trajectory Coherence (Average)	4.30 / 5.0
Latency (Average)	10.24s

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Test Cases Aggregate View



The two failures are isolated to resolution selection, where the system chose a different but operationally defensible action ("RETURN_TO_SENDER" for an ambiguous address, "RESCHEDULE" when the only locker was full).

Shipment	Status	Exception (GT/Pred)	Resolution (GT/Pred)	Escalation (GT/Pred)	Tone (GT/Pred)
SHP-001	DELIVERED	NO / NO	nan / N/A	nan / NO	nan / N/A
SHP-002	ATTEMPTED	YES / YES	RESCHEDULE / RESCHEDULE	YES / YES	FORMAL / FORMAL
SHP-003	ADDRESS_ISSUE	YES / YES	RESCHEDULE / RETURN_TO_SENDER	NO / YES	CASUAL / CASUAL
SHP-004	DAMAGED	YES / YES	REPLACE / REPLACE	YES / YES	FORMAL / FORMAL
SHP-005	ATTEMPTED	YES / YES	REROUTE_TO_LOCKER / RESCHEDULE	YES / YES	CASUAL / CASUAL

Test Cases Aggregate View



Exception identification and tone prediction achieved 100% accuracy across all 10 test cases, indicating that the system never misclassified an exception as noise or vice versa, and tone was always correctly inferred from customer tier.

Shipment	Status	Exception (GT/Pred)	Resolution (GT/Pred)	Escalation (GT/Pred)	Tone (GT/Pred)
SHP-006	REFUSED	YES / YES	RETURN_TO_SENDER / RETURN_TO_SENDER	NO / NO	CASUAL / CASUAL
SHP-007	WEATHER_DELAY	YES / YES	REPLACE / REPLACE	YES / YES	FORMAL / FORMAL
SHP-008	ATTEMPTED	YES / YES	RESCHEDULE / RESCHEDULE	YES / YES	CASUAL / CASUAL
SHP-009	DAMAGED	YES / YES	REPLACE / REPLACE	YES / YES	FORMAL / FORMAL
SHP-010	SCANNED	NO / NO	nan / N/A	nan / NO	nan / N/A

APPENDIX

Detailed Data Description

Table customers, 12 records

Attribute	Type	Values / Description
customer_id	string	Primary key (CUST-001 to CUST-012), Foreign key in delivery logs
name	string	Customer full name (PII)
tier	string	STANDARD, PREMIUM, VIP
preferred_channel	string	SMS, EMAIL
exceptions_last_90d	integer	Range: 0-6
active_credit	float	Range: \$0.00-\$20.00

Detailed Data Description

Table lockers, 6 records

Attribute	Type	Values / Description
locker_id	string	Primary key (LOC-001 to LOC-006)
address	string	Locker street address
zip_code	string	10001-10006
capacity_status	string	AVAILABLE, LIMITED, FULL
operating_hours	string	e.g., "6AM-10PM" or "24 hours"
max_package_size	string	SMALL, MEDIUM, LARGE

Detailed Data Description

CSV file `delivery_logs`, 13 rows, 10 unique shipments

Attribute	Type	Values / Description
shipment_id	string	SHP-001 to SHP-010, Multiple rows may share the same ID for multi-event shipments.
timestamp	datetime	ISO format
status_code	string	One of seven codes: DELIVERED, ATTEMPTED, DAMAGED, ADDRESS_ISSUE, REFUSED, WEATHER_DELAY, SCANNED
status_description	string	Free-text driver/handler notes
customer_id	string	Foreign key to customers table (CUST-001 to CUST-012)
delivery_address	string	Full address including zip code

Detailed Data Description

CSV file `delivery_logs`, 13 rows, 10 unique shipments

Attribute	Type	Values / Description
package_type	string	STANDARD, PERISHABLE, FRAGILE
package_size	string	SMALL, MEDIUM, LARGE
attempt_number	integer	0 for pre-delivery scans, 1+ for delivery attempts
is_duplicate_scan	boolean	True for repeated scan events

Detailed Data Description

CSV file `ground_truth`, 13 rows, aligned 1:1 with delivery logs

Attribute	Type	Values / Description
shipment_id	string	Matches delivery logs (SHP-001 to SHP-010)
is_exception	string	YES, NO
expected_resolution	string	RESCHEDULE, REPLACE, REROUTE_TO_LOCKER, RETURN_TO_SENDER, N/A
expected_tone	string	FORMAL, CASUAL, N/A
should_escalate	string	YES, NO, N/A
ground_truth_reasoning	string	Step-by-step justification tracing each label to specific playbook rules, customer attributes, and locker constraints

Resolution Agent - System Prompt

You are a delivery operations Resolution Agent. Your job is to analyze delivery exceptions and decide the correct resolution action.

You will receive:

- A consolidated delivery event (after deduplication)
- Customer profile (tier, exception history, active credits - no PII)
- Locker availability for the delivery zip code
- Relevant playbook sections
- Escalation rule signals

Your task:

1. Determine if this is a real actionable exception (YES) or noise/routine event (NO).
2. If YES, decide the resolution: RESCHEDULE, REROUTE_TO_LOCKER, REPLACE, or RETURN_TO_SENDER.
3. Provide step-by-step rationale referencing the playbook rules, customer context, and locker constraints.

Exception classification rules:

- No status code is automatically noise. You must read the status_description to determine if the event is routine or exceptional.
- However, these status codes are USUALLY routine and should default to is_exception=NO unless the driver notes indicate something abnormal:
 - DELIVERED: routine unless notes mention damage, wrong recipient, or suspicious circumstances.
 - IN_TRANSIT: routine unless significantly overdue or at an unexpected facility.
 - OUT_FOR_DELIVERY: routine unless notes indicate a problem.
 - SCANNED: routine unless notes indicate a misroute or anomaly.

Resolution Agent - System Prompt

- A simple scan at a depot during normal sorting with no concerning notes is NOT an exception. When in doubt for these status codes, default to NO.

- Duplicate scans (`is_duplicate_scan=True`) are NOT exceptions.

Resolution guidelines:

- RESCHEDULE: Reattempt delivery. Use for failed attempts (1st, 2nd) where redelivery is viable, address issues after customer verification, and weather delays on non-perishable packages. For address issues where the building or street is not found, the correct action is to contact the customer to verify the address and then reschedule - do not default to RETURN_TO_SENDER unless the customer is unresponsive or the address shows clear signs of fraud (vacant lot, demolished building).

- REROUTE_TO_LOCKER: Redirect to a pickup locker. Only if an eligible locker exists (check size, capacity, zip match). Never for perishable packages (no temp control). Avoid for VIP first attempts. If the playbook recommends a locker reroute but no eligible locker is available (all full or size-incompatible), still output REROUTE_TO_LOCKER and note the constraint in your rationale - the Critic will escalate for supervisor intervention.

- REPLACE: Package is compromised. Use for severe damage, leaking/spoiled perishables, and perishable weather delays exceeding 4 hours.

- RETURN_TO_SENDER: Send back to origin. Use ONLY for refused deliveries or confirmed fraud/invalid addresses. Do not use for address issues that can be resolved by contacting the customer.

- Review the `prior_attempt_notes` field if present - earlier driver notes may contain context about access issues, gate codes, or recurring problems that should influence whether to reattempt or reroute.

Consistency rules:

- If `is_exception` is YES, resolution MUST be one of RESCHEDULE, REROUTE_TO_LOCKER, REPLACE, or RETURN_TO_SENDER. Never N/A.

- If `is_exception` is NO, resolution MUST be N/A.

Communication Agent - System Prompt

You are a delivery operations Communication Agent. Your job is to write customer-facing notifications about delivery exceptions.

You will receive:

- The resolution decision and rationale
- Customer name, tier, preferred channel, and active credit
- The original exception details

Your task:

1. Determine the tone based on customer tier:
 - VIP or PREMIUM -> FORMAL (structured, polite, full sentences)
 - STANDARD -> CASUAL (friendly, conversational, brief)
2. Write the notification message including:
 - What happened (plain language, not status codes)
 - What we're doing about it (the resolution)
 - What happens next (timeline, any customer action needed)
 - An apology proportional to severity
3. For EMAIL (VIP/PREMIUM): write a complete email with subject line.
For SMS (STANDARD): keep it under 160 characters if possible, max 300.
4. If the customer has an active service credit, acknowledge prior issues.
5. For locker reroutes: include locker address, operating hours, 3-day pickup deadline, and redelivery option.

You will ALWAYS be invoked for exception cases. You MUST produce both a `tone_label` and a `communication_message`. Never return empty or N/A values - every exception requires a customer notification.

Address the customer by their first name.

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Critic Agent - Resolution Validation Prompt

You are a Critic Agent validating resolution decisions for delivery exceptions.

You will receive the original exception data, customer profile context, locker availability, escalation rule signals, relevant playbook sections, and the Resolution Agent's output.

Check for:

1. Is the `is_exception` classification correct? (DELIVERED, IN_TRANSIT, OUT_FOR_DELIVERY, SCANNED, duplicates = NO)
2. Does the resolution match the playbook rules for this exception type?
3. Were locker constraints respected? (size, capacity, perishable exclusion)
4. Is the rationale logically consistent with the data provided?
5. Were escalation signals acknowledged in the rationale?
6. If `is_exception` is YES but resolution is N/A, return REVISE. These are mutually inconsistent.

CRITICAL ESCALATION RULES:

1. If the escalation signals contain any AUTOMATIC triggers, you MUST return ESCALATE.
2. If `is_exception` == NO, you need not evaluate escalation.

Decide:

- ACCEPT: Resolution is correct and well-reasoned.
- ESCALATE: Resolution may be acceptable but needs supervisor attention due to unaddressed escalation triggers.
- REVISE: Resolution is incorrect or rationale is flawed. Provide specific feedback.

Be strict on playbook compliance. Be lenient on edge cases where the playbook is ambiguous.

Critic Agent - Communication Validation Prompt

You are a Critic Agent validating customer communication for delivery exceptions.

Check for:

1. Tone matches customer tier (VIP/PREMIUM = FORMAL, STANDARD = CASUAL)
2. Message includes: what happened, what we're doing, what happens next, appropriate apology
3. Channel format is correct (EMAIL for VIP/PREMIUM = detailed; SMS for STANDARD = brief)
4. Active service credits referenced if applicable
5. Locker reroute details included if applicable (address, hours, deadline, redelivery option)
6. No status codes exposed to customer, no internal jargon

Decide:

- ACCEPT: Message is appropriate and complete.
- ESCALATE: Message has issues that need supervisor review.

Reasoning Trajectory Coherence Prompt

You are an evaluation judge assessing the reasoning trajectory coherence of a multi-agent delivery exception handling system.

You will receive the full pipeline trace for a single shipment.

Score the reasoning trajectory coherence from 1 to 5:

- 5 - EXCELLENT: Every agent's reasoning logically follows from its inputs. Facts correctly referenced. No contradictions.
- 4 - GOOD: Mostly sound with minor gaps. Correct conclusion despite small shortcuts.
- 3 - ADEQUATE: Right conclusion but partially flawed logic or missed context.
- 2 - POOR: Significant reasoning errors. Agents contradict each other or ignore inputs.
- 1 - INCOHERENT: Reasoning doesn't follow from inputs. Major factual errors.

Respond with ONLY a JSON object: {"score": <1-5>, "justification": "<brief explanation>"}

Test Cases and Observations: Shipment SHP-001

```
--- Predictions ---
Exception: NO (GT: NO)
Resolution: N/A (GT: nan)
Escalated: NO (GT: nan)
Tone:      N/A (GT: nan)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---
Task Complete: PASS
Exception ID:  PASS
Resolution:    PASS
Tone:          PASS
Escalation Accuracy: N/A
Tool Call Accuracy: PASS
Coherence Score: 5/5
Latency:        0.01s

--- Trajectory ---
preprocessor: 1 raw rows -> 1 after dedup
preprocessor: DELIVERED flagged as noise by guardrail, skipping tool calls
orchestrator: Noise override from preprocessor, skipping to finalize
finalize: actions={"shipment_id": "SHP-001", "is_exception": "NO", "resolution": "N/A", "escalated": false, "tone": "N/A",
"message": "", "revision_count": 0, "guardrail_blocked": false}; latency=0.006s

--- Document Citations ---
Playbook pages referenced: None
```

The Successful Delivery (Noise Filtering) Case successfully showcases how the system short-circuits "DELIVERED" scans by bypassing all tool, RAG, and LLM invocations to achieve near-zero latency and correct "NO/N/A" resolution.

Test Cases and Observations: Shipment SHP-002

--- Predictions ---

Exception: YES (GT: YES)
Resolution: RESCHEDULE (GT: RESCHEDULE)
Escalated: YES (GT: YES)
Tone: FORMAL (GT: FORMAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: PASS
Exception ID: PASS
Resolution: PASS
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 4/5
Latency: 13.75s

--- Trajectory ---

preprocessor: 3 raw rows -> 2 after dedup
resolution_agent: is_exception=YES, resolution=RESCHEDULE
critic_resolution: decision=ESCALATE
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: VIP customer with 5 exceptions in 90d (>=3)']
communication_agent: tone=FORMAL
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: VIP customer with 5 exceptions in 90d (>=3)']
critic_communication: decision=ESCALATE
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: VIP customer with 5 exceptions in 90d (>=3)']

Test Cases and Observations: Shipment SHP-002

```
finalize: actions={"shipment_id": "SHP-002", "is_exception": "YES", "resolution": "RESCHEDULE", "escalated": true, "tone": "FORMAL",  
"message": "Subject: Delivery Update for Your Package\n\nDear John Smith,\n\nI hope this message finds you well. I am writing to  
inform you about the recent delivery attempt for your package. Unfortunately, our driver attempted to deliver your package but was  
unable to do so as no one was home. A second notice was left, and we understand this can be inconvenient.\n\nTo resolve this, we  
would like to reschedule the delivery for a time that is more convenient for you. Please let us know if you will be available for a  
third attempt or if you would prefer to discuss alternative options.\n\nWe appreciate your understanding and apologize for any  
inconvenience this may have caused. Additionally, we acknowledge your active service credit of $15.00 due to prior issues, and we  
are committed to improving your experience.\n\nThank you for your attention to this matter. We look forward to your  
response.\n\nBest regards,\n\n[Your Name]  \nDelivery Operations Team", "revision_count": 0, "guardrail_blocked": false};  
latency=13.747s
```

--- Document Citations ---

Playbook pages referenced: Page 5, Page 5, Page 2, Page 2

--- Customer Message ---

Subject: Delivery Update for Your Package

Dear John Smith,

I hope this message finds you well. I am writing to inform you about the recent delivery attempt for your package. Unfortunately, our drive...

The VIP Multi-Attempt with Duplicate Scan Case validates the system's "defense-in-depth" architecture by successfully deduplicating scan events, applying VIP-specific business rules, and generating a PII-personalized formal resolution that adheres to the established playbook.

Test Cases and Observations: Shipment SHP-003

--- Predictions ---

Exception: YES (GT: YES)
Resolution: RETURN_TO_SENDER (GT: RESCHEDULE)
Escalated: YES (GT: NO)
Tone: CASUAL (GT: CASUAL)
Revisions: 2
Guardrail: CLEAR

--- Metrics ---

Task Complete: FAIL
Exception ID: PASS
Resolution: FAIL
Tone: PASS
Escalation Accuracy: FAIL
Tool Call Accuracy: PASS
Coherence Score: 3/5
Latency: 19.24s

--- Trajectory ---

preprocessor: 1 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=RESCHEDULE
critic_resolution: decision=REVISE
orchestrator: REVISE loop 1/2
resolution_agent: is_exception=YES, resolution=RETURN_TO_SENDER
critic_resolution: decision=REVISE
orchestrator: Max loops reached, forcing ESCALATE
communication_agent: tone=CASUAL

Test Cases and Observations: Shipment SHP-003

```
communication_agent: tone=CASUAL
critic_communication: decision=ACCEPT
finalize: actions={"shipment_id": "SHP-003", "is_exception": "YES", "resolution": "RETURN_TO_SENDER", "escalated": true, "tone":
"CASUAL", "message": "Hey David! We couldn't find your building at the address provided. We'll return your package to the sender. If
you have a new address, let us know within 48 hours!", "revision_count": 2, "guardrail_blocked": false}; latency=19.244s
```

--- Document Citations ---

Playbook pages referenced: Page 3, Page 3, Page 2, Page 3

--- Customer Message ---

Hey David! We couldn't find your building at the address provided. We'll return your package to the sender. If you have a new address, let us know within 48 hours!

The Address Not Found (Standard Customer) Case highlights a "successful failure" where the Critic-driven Revision Loop exhausted its retry limit on a borderline driver note, ultimately triggering a safe ESCALATE fallback that prioritizes human oversight over an uncertain autonomous decision.

Test Cases and Observations: Shipment SHP-004

--- Predictions ---

Exception: YES (GT: YES)
Resolution: REPLACE (GT: REPLACE)
Escalated: YES (GT: YES)
Tone: FORMAL (GT: FORMAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: PASS
Exception ID: PASS
Resolution: PASS
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 5/5
Latency: 13.15s

--- Trajectory ---

preprocessor: 1 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=REPLACE
critic_resolution: decision=ESCALATE
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: VIP customer with 4 exceptions in 90d (>=3)']
communication_agent: tone=FORMAL
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: VIP customer with 4 exceptions in 90d (>=3)']
critic_communication: decision=ESCALATE

Test Cases and Observations: Shipment SHP-004

```
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: VIP customer with 4 exceptions in 90d (>=3)']
finalize: actions={"shipment_id": "SHP-004", "is_exception": "YES", "resolution": "REPLACE", "escalated": true, "tone": "FORMAL",
"message": "Subject: Important Update Regarding Your Recent Delivery\n\nDear Emily Davis,\n\nI hope this message finds you well. I
am writing to inform you about an issue with your recent delivery. Unfortunately, the package containing your fragile item was
damaged during transit, resulting in a crushed box with a visible dent. We understand how important this item is to you, and we
sincerely apologize for any inconvenience this may cause.\n\nTo resolve this matter, we will be initiating a replacement order for
you. Given the nature of the damage, we believe this is the best course of action to ensure you receive your item in perfect
condition.\n\nYou can expect the replacement to be processed shortly, and we will keep you updated on the timeline for delivery. In
the meantime, please rest assured that we are taking all necessary steps to prevent such occurrences in the future.\n\nThank you for
your understanding and patience. If you have any questions or require further assistance, please do not hesitate to reach
out.\n\nWarm regards,\n\n[Your Name] \nDelivery Operations Team", "revision_count": 0, "guardrail_blocked": false}; latency=13.150s
```

--- Document Citations ---

Playbook pages referenced: Page 4, Page 4, Page 4, Page 6

--- Customer Message ---

Subject: Important Update Regarding Your Recent Delivery

Dear Emily Davis,

I hope this message finds you well. I am writing to inform you about an issue with your recent delivery. Unfortunately, the...

The Damaged Fragile Package (VIP Escalation) Case demonstrates the system's ability to layer complex business logic—automatically lowering the "REPLACE" threshold for fragile items while simultaneously triggering a double-lock escalation (Critic + Rule Engine) to ensure a high-touch human handoff for a VIP customer.

Test Cases and Observations: Shipment SHP-005

--- Predictions ---

Exception: YES (GT: YES)
Resolution: RESCHEDULE (GT: REROUTE_TO_LOCKER)
Escalated: YES (GT: YES)
Tone: CASUAL (GT: CASUAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: FAIL
Exception ID: PASS
Resolution: FAIL
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 2/5
Latency: 10.01s

--- Trajectory ---

preprocessor: 1 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=RESCHEDULE
critic_resolution: decision=ESCALATE
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: 3rd failed delivery attempt']
communication_agent: tone=CASUAL
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: 3rd failed delivery attempt']
critic_communication: decision=ESCALATE

Test Cases and Observations: Shipment SHP-005

```
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: 3rd failed delivery attempt']
finalize: actions={"shipment_id": "SHP-005", "is_exception": "YES", "resolution": "RESCHEDULE", "escalated": true, "tone": "CASUAL",
"message": "Hey Thomas! We tried delivering your package but couldn't reach you after 3 attempts. Since your building doesn't have a
buzzer, we'll need to reschedule. Let us know a good time for you!", "revision_count": 0, "guardrail_blocked": false};
latency=10.010s
```

--- Document Citations ---

Playbook pages referenced: Page 2, Page 3, Page 3, Page 5

--- Customer Message ---

Hey Thomas! We tried delivering your package but couldn't reach you after 3 attempts. Since your building doesn't have a buzzer, we'll need to reschedule. Let us know a good time for you!

The Third Failed Attempt (Locker Full, Mandatory Escalation) Case demonstrates a divergence between "Ground Truth" (ideal playbook path) and "Operational Reality" (available capacity), where the system intelligently bypassed a full locker (LOC-003) in favor of a feasible RESCHEDULE action.

Test Cases and Observations: Shipment SHP-006

--- Predictions ---

Exception: YES (GT: YES)
Resolution: RETURN_TO_SENDER (GT: RETURN_TO_SENDER)
Escalated: NO (GT: NO)
Tone: CASUAL (GT: CASUAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: PASS
Exception ID: PASS
Resolution: PASS
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 5/5
Latency: 8.82s

--- Trajectory ---

preprocessor: 1 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=RETURN_TO_SENDER
critic_resolution: decision=ACCEPT
communication_agent: tone=CASUAL
critic_communication: decision=ACCEPT

Test Cases and Observations: Shipment SHP-006

```
finalize: actions={"shipment_id": "SHP-006", "is_exception": "YES", "resolution": "RETURN_TO_SENDER", "escalated": false, "tone": "CASUAL", "message": "Hey Michael! Your package was refused at delivery since you mentioned you didn\u2019t order it. We\u2019ll return it to the sender within 2 business days. No action needed from you!", "revision_count": 0, "guardrail_blocked": false}; latency=8.822s
```

--- Document Citations ---

Playbook pages referenced: Page 5, Page 5, Page 5, Page 2

--- Customer Message ---

Hey Michael! Your package was refused at delivery since you mentioned you didn't order it. We'll return it to the sender within 2 business days. No action needed from you!

The Refused Delivery (Return to Sender) Case represents the system's "Golden Path" for exceptions, achieving high efficiency and a "Clean Pass" by navigating unambiguous playbook rules for a non-premium customer without unnecessary escalation or revision.

Test Cases and Observations: Shipment SHP-007

--- Predictions ---

Exception: YES (GT: YES)
Resolution: REPLACE (GT: REPLACE)
Escalated: YES (GT: YES)
Tone: FORMAL (GT: FORMAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: PASS
Exception ID: PASS
Resolution: PASS
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 5/5
Latency: 12.67s

--- Trajectory ---

preprocessor: 2 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=REPLACE
critic_resolution: decision=ESCALATE
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: Perishable with 5.0hr delay (>4hr threshold)']
communication_agent: tone=FORMAL
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: Perishable with 5.0hr delay (>4hr threshold)']
critic_communication: decision=ACCEPT
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: Perishable with 5.0hr delay (>4hr threshold)']

Test Cases and Observations: Shipment SHP-007

```
finalize: actions={"shipment_id": "SHP-007", "is_exception": "YES", "resolution": "REPLACE", "escalated": true, "tone": "FORMAL",  
"message": "Subject: Update on Your Delivery - Weather Delay\n\nDear Maria Garcia,\n\nI hope this message finds you well. I am  
writing to inform you about a delay in the delivery of your recent order due to heavy snow that has blocked the roads, resulting in  
an estimated delay of 5 hours.\n\nGiven that your package contains perishable items, we have determined that this delay exceeds the  
4-hour threshold for safe delivery. Therefore, we will be replacing your order to ensure you receive your items in optimal  
condition.\n\nYou do not need to take any action at this time. We will process the replacement and notify you once it is on its way.  
We sincerely apologize for any inconvenience this may cause, especially considering your active service credit from previous  
issues.\n\nThank you for your understanding and patience during this time. If you have any questions or need further assistance,  
please do not hesitate to reach out.\n\nBest regards,\n\n[Your Name]  \nDelivery Operations Team", "revision_count": 0,  
"guardrail_blocked": false}; latency=12.668s
```

--- Document Citations ---

Playbook pages referenced: Page 6, Page 6, Page 6, Page 6

--- Customer Message ---

Subject: Update on Your Delivery - Weather Delay

Dear Maria Garcia,

I hope this message finds you well. I am writing to inform you about a delay in the delivery of your recent order due to heavy
sno...

The Perishable Weather Delay Exceeding Threshold (VIP Replace) Case serves as a textbook example of the system's Deterministic Authority, where regex-parsed data ("5hr") triggered a hard-coded replacement and escalation rule that overrode any potential LLM ambiguity regarding perishable thresholds.

Test Cases and Observations: Shipment SHP-008

--- Predictions ---

Exception: YES (GT: YES)
Resolution: RESCHEDULE (GT: RESCHEDULE)
Escalated: YES (GT: YES)
Tone: CASUAL (GT: CASUAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: PASS
Exception ID: PASS
Resolution: PASS
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 4/5
Latency: 13.95s

--- Trajectory ---

preprocessor: 1 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=RESCHEDULE
critic_resolution: decision=ESCALATE
communication_agent: tone=CASUAL
critic_communication: decision=ACCEPT

Test Cases and Observations: Shipment SHP-008

```
finalize: actions={"shipment_id": "SHP-008", "is_exception": "YES", "resolution": "RESCHEDULE", "escalated": true, "tone": "CASUAL",  
"message": "Hey Thomas! We tried delivering your package but you weren't home. No worries, we'll reschedule it for the next  
business day. Thanks for your patience!", "revision_count": 0, "guardrail_blocked": false}; latency=13.950s
```

--- Document Citations ---

Playbook pages referenced: Page 2, Page 3, Page 5, Page 3

--- Customer Message ---

Hey Thomas! We tried delivering your package but you weren't home. No worries, we'll reschedule it for the next business day. Thanks for your patience!

The Discretionary Escalation (High Exception History) Case highlights the Critic's Discretionary Power, where the model successfully identified a systemic delivery failure (6 exceptions in 90 days) and triggered a manual escalation despite no hard-coded rules being met.

Test Cases and Observations: Shipment SHP-009

--- Predictions ---

Exception: YES (GT: YES)
Resolution: REPLACE (GT: REPLACE)
Escalated: YES (GT: YES)
Tone: FORMAL (GT: FORMAL)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---

Task Complete: PASS
Exception ID: PASS
Resolution: PASS
Tone: PASS
Escalation Accuracy: PASS
Tool Call Accuracy: PASS
Coherence Score: 5/5
Latency: 10.76s

--- Trajectory ---

preprocessor: 1 raw rows -> 1 after dedup
resolution_agent: is_exception=YES, resolution=REPLACE
critic_resolution: decision=ESCALATE
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: Damaged perishable package']
communication_agent: tone=FORMAL
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: Damaged perishable package']
critic_communication: decision=ACCEPT
orchestrator: Forced escalation from rule engine - ['AUTOMATIC: Damaged perishable package']

Test Cases and Observations: Shipment SHP-009

```
finalize: actions={"shipment_id": "SHP-009", "is_exception": "YES", "resolution": "REPLACE", "escalated": true, "tone": "FORMAL",  
"message": "Subject: Update on Your Recent Delivery Issue\n\nDear Sarah Brown,\n\nI hope this message finds you well. I am writing  
to inform you about an issue with your recent delivery. Unfortunately, your package was damaged during transit, resulting in a leak  
and an unpleasant odor, likely due to spoiled food items. \n\nWe understand how important it is for you to receive your orders in  
perfect condition, especially as a valued PREMIUM member. Therefore, we have initiated a replacement order for your perishable  
items, as per our guidelines for severe damage.\n\nYou can expect the replacement to be dispatched shortly, and we will keep you  
updated on the timeline for delivery. We sincerely apologize for any inconvenience this may have caused, especially considering your  
active service credit of $5.00 from previous issues.\n\nThank you for your understanding and patience. If you have any further  
questions or concerns, please do not hesitate to reach out.\n\nBest regards,\n\n[Your Name] \nDelivery Operations Team",  
"revision_count": 0, "guardrail_blocked": false}; latency=10.764s
```

--- Document Citations ---

Playbook pages referenced: Page 4, Page 6, Page 4, Page 10

--- Customer Message ---

Subject: Update on Your Recent Delivery Issue

Dear Sarah Brown,

I hope this message finds you well. I am writing to inform you about an issue with your recent delivery. Unfortunately, your package
w...

The Severely Damaged Perishable (Premium Replace) Case represents a high-integrity execution where the system correctly identified an "undeliverable" state, triggering immediate replacement and redundant escalation for a PREMIUM customer.

Test Cases and Observations: Shipment SHP-010

```
--- Predictions ---
Exception: NO (GT: NO)
Resolution: N/A (GT: nan)
Escalated: NO (GT: nan)
Tone:      N/A (GT: nan)
Revisions: 0
Guardrail: CLEAR

--- Metrics ---
Task Complete: PASS
Exception ID:  PASS
Resolution:    PASS
Tone:          PASS
Escalation Accuracy: N/A
Tool Call Accuracy: PASS
Coherence Score: 5/5
Latency:        0.01s

--- Trajectory ---
preprocessor: 1 raw rows -> 1 after dedup
preprocessor: SCANNED flagged as noise by guardrail, skipping tool calls
orchestrator: Noise override from preprocessor, skipping to finalize
finalize: actions={"shipment_id": "SHP-010", "is_exception": "NO", "resolution": "N/A", "escalated": false, "tone": "N/A",
"message": "", "revision_count": 0, "guardrail_blocked": false}; latency=0.005s

--- Document Citations ---
Playbook pages referenced: None
```

The Routine Depot Scan (Noise Filtering) Case confirms the system's ability to maintain high-efficiency "Noise" handling across varying status codes (SCANNED vs. DELIVERED), ensuring that non-exception events never reach the expensive LLM or RAG layers.



Power Ahead!

